

Benchmarking Lightweight CNN for Real-Time Classification of Bangladeshi Mango Varieties on Edge Devices

Undergraduate Thesis Defense

Computer Science and Engineering
Jahangirnagar University

Presentation Outline

Part I: Foundation

- Introduction & Background
- Problem Statement
- Research Objectives
- Literature Review Summary

Part II: Implementation

- Proposed Framework & Dataset
- Benchmarking CNN Architectures
- AmmChini AI App Deployment
- Conclusion & Future Work

Introduction

- Mango is an economically significant horticultural crop in Bangladesh.
- Accurate identification is essential for pricing, quality control, and export chains.
- Current identification methods rely strictly on manual visual inspection.
- Manual sorting is labor-intensive, subjective, and highly inconsistent.
- Strong need for automated, objective classification systems in rural orchards.



Background & Motivation



High Similarity

Indigenous varieties like Langra and Kanchon Langra exhibit extremely subtle morphological differences.



Deep Learning

CNNs can automatically learn discriminative visual features, vastly outperforming manual human extraction.



Field Deployment

Real-world agriculture requires portable, reliable systems accessible to farmers directly in the orchard.

Problem Statement



Heavy Computational Demands: State-of-the-art CNNs (ResNet, DenseNet) require powerful GPUs, rendering them unsuitable for low-end mobile edge devices.



Cloud Dependency: Transmitting images to servers causes latency and entirely fails in rural areas lacking stable internet infrastructure.



Fine-Grained Classification: Existing models often struggle with the subtle inter-class similarities unique to Bangladeshi mangoes.



Deployment Gap: A distinct lack of end-to-end frameworks that balance predictive performance with realistic mobile efficiency.

Research Objectives

- 🎯 **Benchmark Architectures:** Evaluate 5 lightweight CNNs on the Mangifera2012 dataset for performance and efficiency.
- 🎯 **Architecture-Specific Preprocessing:** Implement robust, model-specific training pipelines for fair comparative evaluation.
- 🎯 **Optimize for Edge:** Convert the best-performing model to TensorFlow Lite using Float16 post-training quantization.
- 🎯 **Mobile Deployment:** Develop "AmmChini AI", a Flutter app for fully offline, real-time prediction.

Literature Review Summary

Current Trends

Most studies prioritize accuracy using computationally expensive models (e.g., ResNet50, YOLOv4).

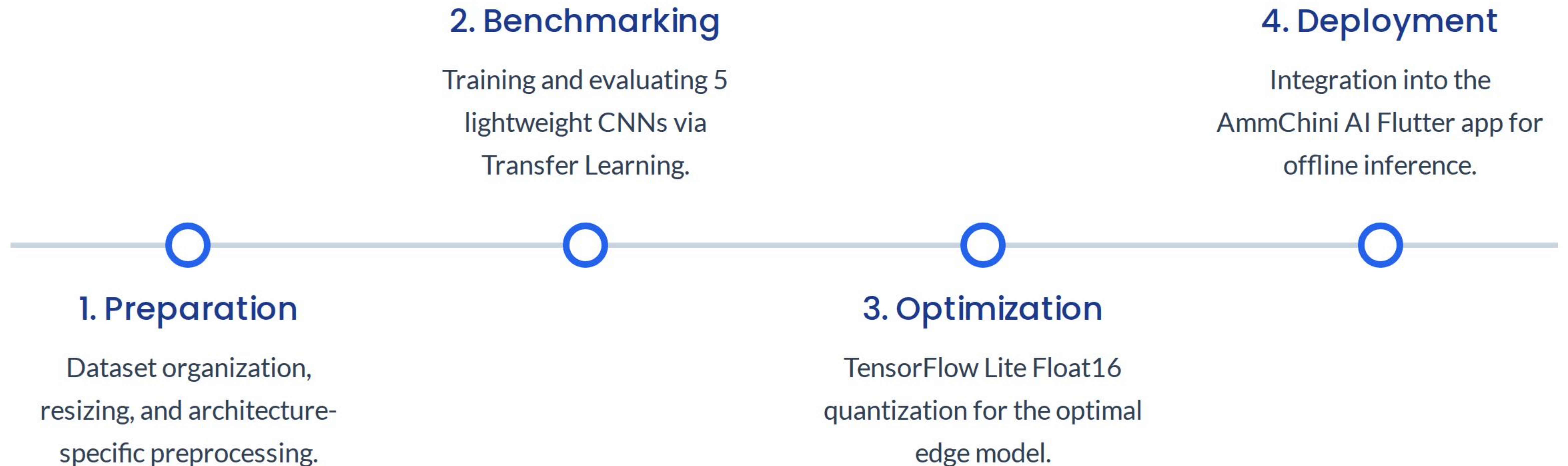
Datasets primarily feature international commercial cultivars (Kent, Alphonso) rather than indigenous ones.

Identified Research Gap

Limited benchmarking of lightweight models strictly for resource-constrained mobile deployment.

Lack of complete end-to-end Edge AI implementations capable of functioning without internet connectivity.

Proposed Framework



Dataset Description

Mangifera2012 Dataset: 2,012 High-Resolution RGB Images

Variety	Image Count	Variety	Image Count
Amrapali	252	Kanchon Langra	210
Bari-4	235	Katimon	163
Bari-7	176	Langra	202
Fazlee	156	Mollika	221
Harivanga	202	Nilambori	195

Preprocessing & Augmentation



Image Resizing

Images standardized to 224x224 pixels to balance computational efficiency and fine-grained feature preservation.



Dynamic Preprocessing

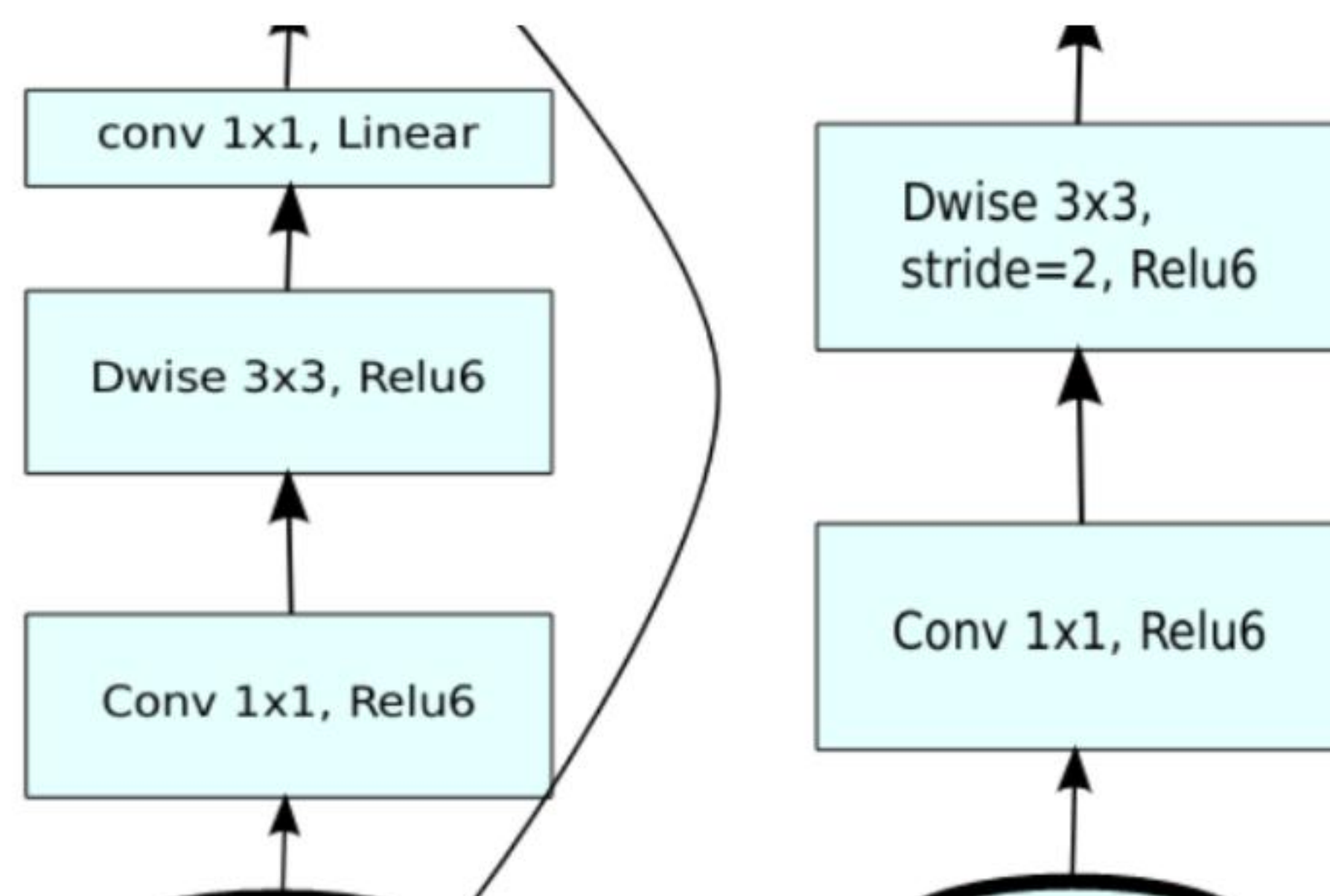
Architecture-specific normalization applied dynamically to match ImageNet pre-training distributions.



Online Augmentation

Random rotation, shift, shear, zoom, and flip applied dynamically during training to enhance generalization.

Benchmarked CNN Architectures



MobileNetV2 & V3-Small:

Inverted residuals, depthwise separable convolution, Squeeze-and-Excitation.

Abstract CNN Architecture

EfficientNetV2B0 & DenseNet121:

Compound scaling, fused MBConv blocks, and dense feature reuse.



NASNetMobile:

Automatically discovered optimal cell structures using Neural Architecture Search.

Experimental Setup



Transfer Learning: All models initialized with ImageNet pre-trained weights to leverage generic visual features.



Frozen Backbone: Base feature extractors were frozen during initial training to prevent immediate overfitting.



Custom Classification Head: The exact same custom head was appended to all models to ensure fair comparison:

- Global Average Pooling Layer
- Dropout Layer (0.30 Rate)
- Dense Layer (128 neurons, ReLU Activation)
- Softmax Output Layer (10 classes)

Performance Comparison

CNN Model	Accuracy (%)	Weighted F1 (%)	Size (MB)	Latency (ms)
EfficientNetV2B0	92.00	91.98	24.6	57
MobileNetV3-Small	91.25	91.22	7.2	31
DenseNet121	90.25	90.21	33.1	71
MobileNetV2	89.75	89.71	9.8	43
NASNetMobile	88.50	88.42	20.5	49

* MobileNetV3-Small selected for optimal Edge AI deployment balance.

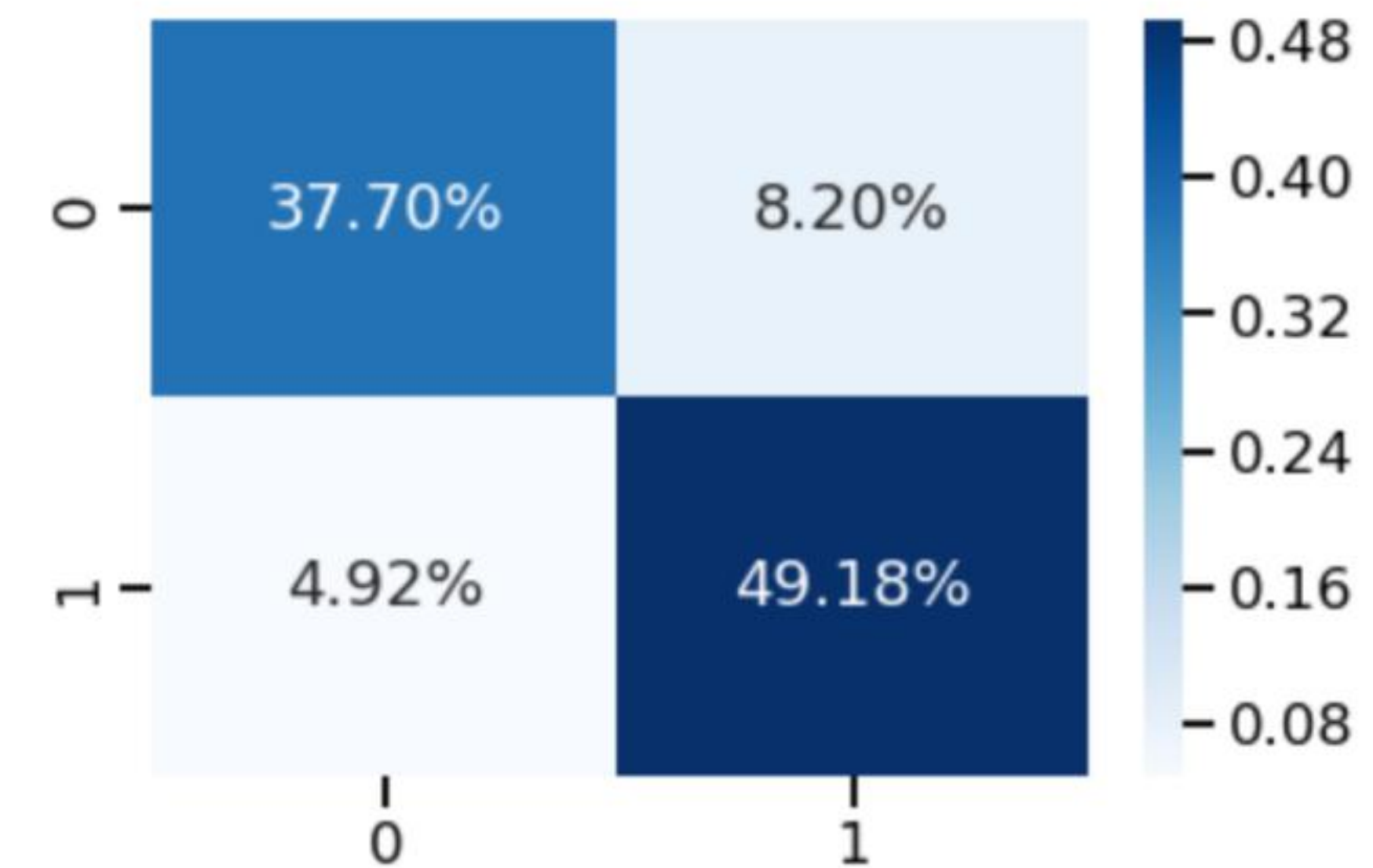
Evaluating Model Efficiency



MobileNetV3-Small achieves near-parity in accuracy while remaining a fraction of the physical size, demonstrating extreme parameter efficiency.

Confusion Matrix Analysis

- **Perfect Classification:** Fazlee, Katimon, and Mollika achieved 100% recognition without error.
- **High Accuracy:** Amrapali and Langra demonstrated excellent separability.
- **Significant Improvement:** MobileNetV3-Small resolved much of the Bari-4 vs. Kanchon Langra confusion seen in baseline models.
- **Persisting Challenge:** Nilambori remained difficult, often confused with Bari-7 and Bari-4 due to near-identical green peel textures.



Why MobileNetV3-Small?

31ms

Inference Latency

The Optimal Edge Choice

While EfficientNet achieved slightly higher accuracy, it was too heavy for low-end smartphones. MobileNetV3-Small was chosen because it provides:

- **Storage:** Only 7.2 MB model size.
- **Speed:** Real-time execution without delays.
- **Accuracy:** 91.25% robust precision and recall.
- **Architecture:** Hard-Swish and Squeeze-and-Excitation mechanisms ideal for fine-grained Edge AI.

TensorFlow Lite Quantization

Float16 Post-Training Quantization

To enable highly efficient smartphone execution, the selected model was converted to TFLite format.

- Reduces model weights from 32-bit floats to 16-bit floats.
- Drastically shrinks storage footprint on the device.
- Maintains identical classification performance.
- Lowers CPU memory consumption during active inference.



AmmChini AI App Architecture

Flutter Frontend

Cross-platform UI handling the camera interface and gallery image selection.

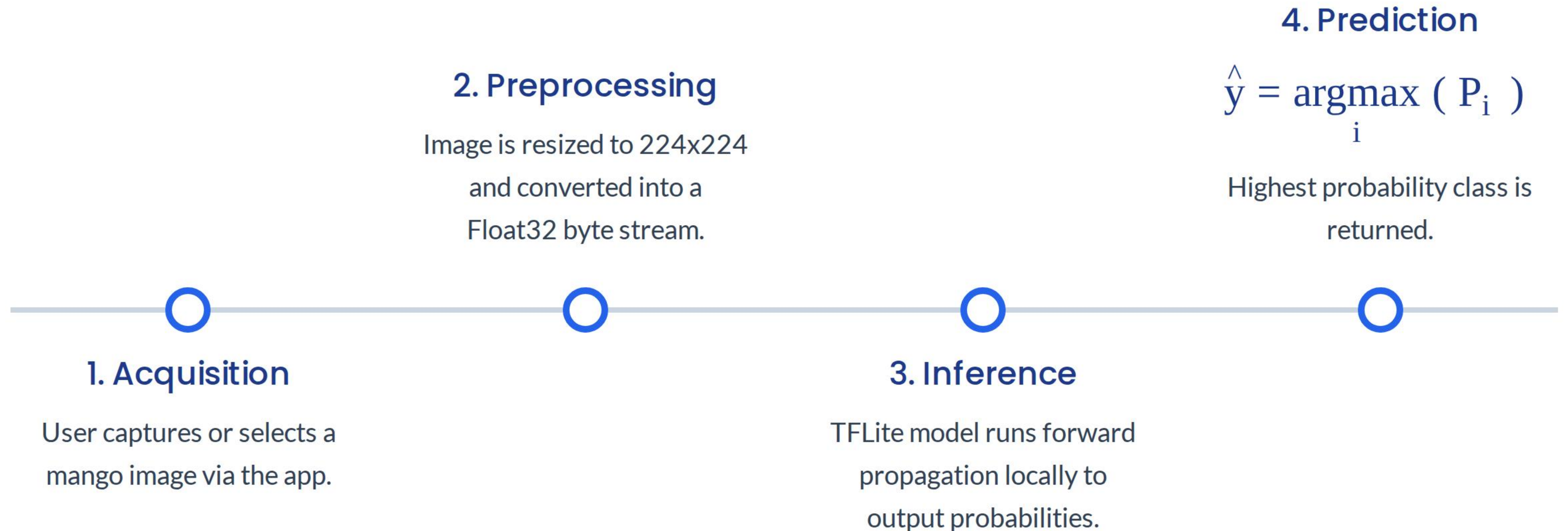
Responsive design tailored for rural users to view predictions clearly and easily.

TFLite Backend

Local execution of the MobileNetV3 interpreter directly on the device CPU.

Zero reliance on external APIs or servers, ensuring complete offline capability.

Live Prediction Workflow



Advantages of Edge AI



Offline Operation

Works seamlessly in rural orchards lacking reliable internet or cellular infrastructure.



Low Latency

Sub-second, real-time inference without frustrating network communication delays.



Privacy & Cost

Images never leave the device, ensuring privacy and eliminating recurring cloud API costs.

Conclusion

This research successfully bridged the gap between academic computer vision and practical agriculture. By benchmarking and optimizing MobileNetV3-Small, we delivered **AmmChini AI**: a highly accurate, fully offline Edge AI solution for classifying indigenous Bangladeshi mango varieties.

Future Work



Dataset Expansion: Collect more images across varying geographical regions, lighting, and maturity stages.



Object Detection: Transition from single-image classification to bounding-box detection for batch sorting.



Disease Diagnosis: Integrate multi-task learning to detect plant diseases alongside variety classification.



Explainable AI (XAI): Implement techniques like Grad-CAM to show farmers exactly which features influenced the AI's decision.

Questions & Answers

Thank you for your attention and time.

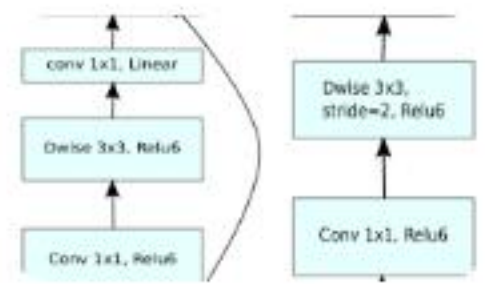
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Image Sources



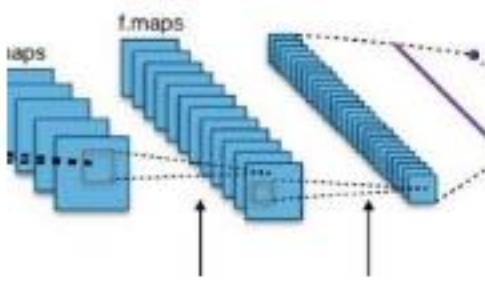
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Source: mmpretrain.readthedocs.io



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Source: www.edgectortex.com